B5. Segurança e controle de acesso – Testes e Evidências

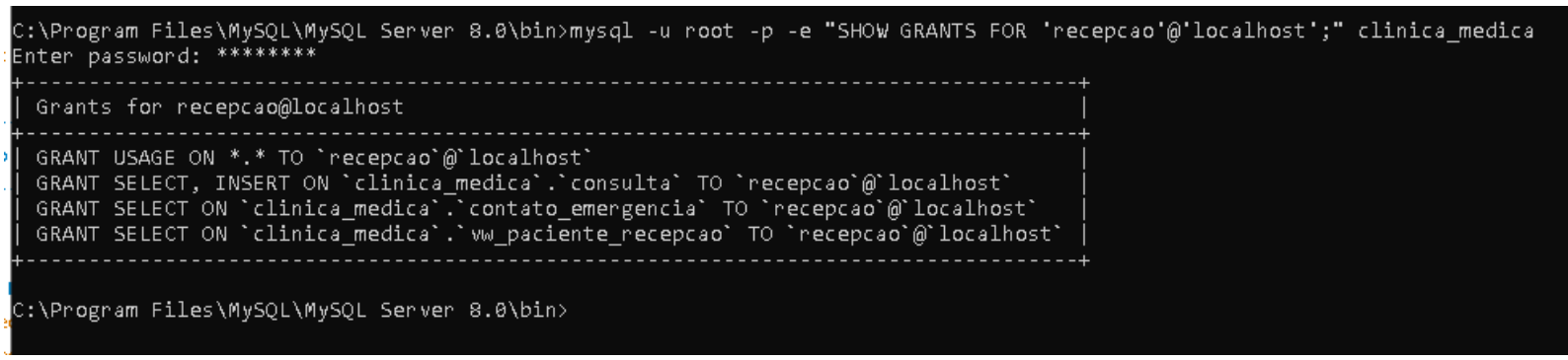
As imagens a seguir apresentam as evidências de execução dos testes de segurança e em todos eles o resultado obtido coincidiu com o resultado esperado. Alguns perfis já foram logados com senha para facilitar o processo.

1. Prova de previlégios (root)

Teste 1.1

mysql -u root -p -e "SHOW GRANTS FOR 'recepcao'@'localhost';" clinica_medica

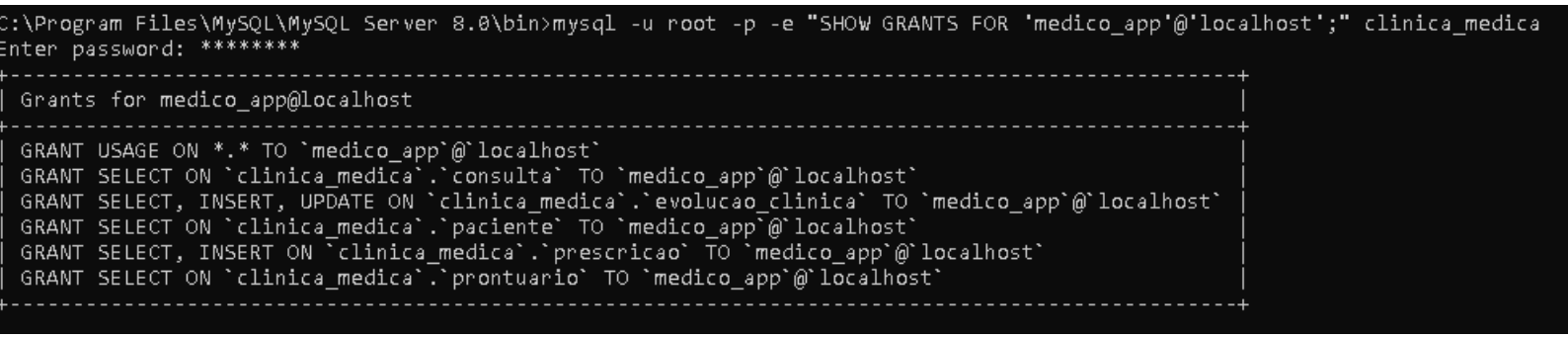
Evidência:



Teste 1.2

mysql -u root -p -e "SHOW GRANTS FOR 'medico_app'@'localhost';" clinica_medica

Evidência:



Teste 1.3

mysql -u root -p -e "SHOW GRANTS FOR 'admin_clinica'@'localhost';" clinica_medica

Evidência:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p -e "SHOW GRANTS FOR 'admin_clinica'@'localhost';" clinica_medica
Enter password: ********
+-----------------------------------------------------------------------------------------------------------------+
| Grants for admin_clinica@localhost                                                                              |
+-----------------------------------------------------------------------------------------------------------------+
| GRANT USAGE ON *.* TO `admin_clinica`@`localhost`                                                               |
| GRANT SELECT, INSERT, UPDATE, DELETE ON `clinica_medica`.`consulta` TO `admin_clinica`@`localhost`              |
| GRANT SELECT, INSERT, UPDATE, DELETE ON `clinica_medica`.`contato_emergencia` TO `admin_clinica`@`localhost`    |
| GRANT SELECT, INSERT, UPDATE, DELETE ON `clinica_medica`.`enfermeiro` TO `admin_clinica`@`localhost`            |
| GRANT SELECT, INSERT, UPDATE, DELETE ON `clinica_medica`.`especialidade` TO `admin_clinica`@`localhost`         |
| GRANT SELECT ON `clinica_medica`.`evolucao_clinica` TO `admin_clinica`@`localhost`                              |
| GRANT SELECT, INSERT, UPDATE, DELETE ON `clinica_medica`.`medicamento` TO `admin_clinica`@`localhost`           |
| GRANT SELECT, INSERT, UPDATE, DELETE ON `clinica_medica`.`medico_especialidade` TO `admin_clinica`@`localhost` |
| GRANT SELECT, INSERT, UPDATE, DELETE ON `clinica_medica`.`medico` TO `admin_clinica`@`localhost`                |
| GRANT SELECT, INSERT, UPDATE, DELETE ON `clinica_medica`.`paciente` TO `admin_clinica`@`localhost`              |
| GRANT SELECT ON `clinica_medica`.`prescricao` TO `admin_clinica`@`localhost`                                    |
| GRANT SELECT, INSERT, UPDATE, DELETE ON `clinica_medica`.`profissional_saude` TO `admin_clinica`@`localhost`    |
| GRANT SELECT ON `clinica_medica`.`prontuario` TO `admin_clinica`@`localhost`                                    |
+-----------------------------------------------------------------------------------------------------------------+
```

2. Prova da visão (recepção)

Teste 2.1

mysql -u recepcao -pR3c3p!2026#Clin -e "SELECT * FROM vw_paciente_recepcao LIMIT 3;" clinica_medica

Evidência:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u recepcao -p -e "SELECT * FROM vw_paciente_recepcao LIMIT 3;" clinica_medica
Enter password: ***************
+-------------+-----------------+---------------+-----------------+
| id_paciente | nome            | cpf_mascarado | data_nascimento |
+-------------+-----------------+---------------+-----------------+
|           1 | Arthur Monteiro | ***.***.***-61 | 1998-02-14      |
|           2 | Helena Duarte   | ***.***.***-52 | 2001-07-23      |
|           3 | Bruno Tavares   | ***.***.***-27 | 1987-11-05      |
+-------------+-----------------+---------------+-----------------+
```

Teste 2.2

mysql -u recepcao -pR3c3p!2026#Clin -e "SELECT * FROM Paciente LIMIT 1;" clinica_medica

Evidência:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u recepcao -pR3c3p!2026#Clin -e "SELECT * FROM Paciente LIMIT 1;" clinica_medica
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1142 (42000) at line 1: SELECT command denied to user 'recepcao'@'localhost' for table 'paciente'

C:\Program Files\MySQL\MySQL Server 8.0\bin>
```

Teste 2.3

mysql -u recepcao -pR3c3p!2026#Clin -e "SELECT * FROM Prontuario LIMIT 1;" clinica_medica

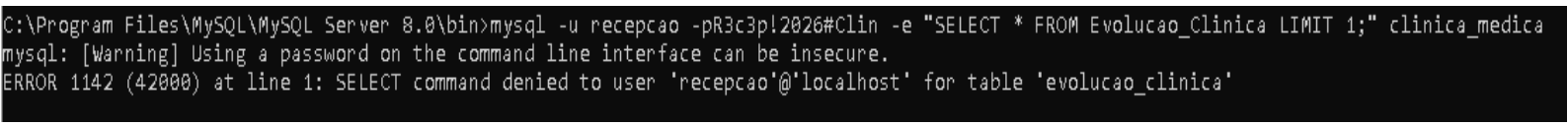
Evidência:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u recepcao -pR3c3p!2026#Clin -e "SELECT * FROM Prontuario LIMIT 1;" clinica_medica
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1142 (42000) at line 1: SELECT command denied to user 'recepcao'@'localhost' for table 'prontuario'
```

Teste 2.4

mysql -u recepcao -pR3c3p!2026#Clin -e "SELECT * FROM Evolucao_Clinica LIMIT 1;" clinica_medica

Evidência:



3. Prova de menor previlégio (médico e admin)

Teste 3.1

mysql -u medico_app -pM3d!c0_2026#App -e "DELETE FROM Paciente WHERE id_paciente = 1;" clinica_medica

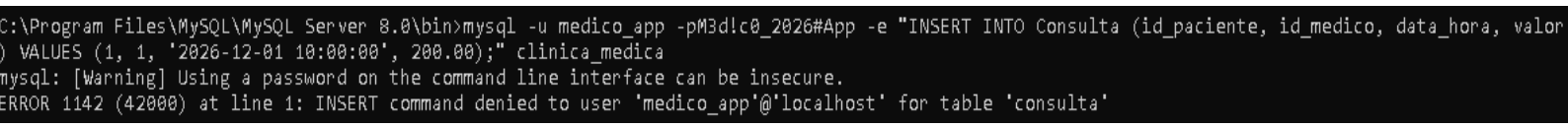
Evidência:



Teste 3.2

mysql -u medico_app -pM3d!c0_2026#App -e "INSERT INTO Consulta (id_paciente, id_medico, data_hora, valor) VALUES (1, 1, '2026-12-01 10:00:00', 200.00);" clinica_medica

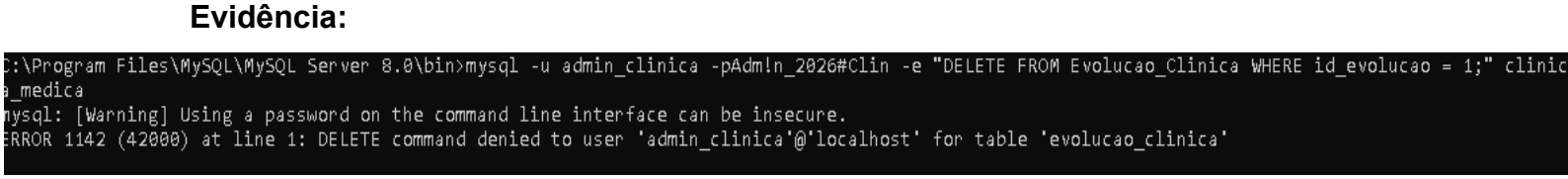
Evidência:



Teste 3.3

mysql -u admin_clinica -pAdm!n_2026#Clin -e "DELETE FROM Evolucao_Clinica WHERE id_evolucao = 1;" clinica_medica

Evidência:



Teste 3.4

mysql -u admin_clinica -pAdm!n_2026#Clin -e "DELETE FROM Prontuario WHERE id_prontuario = 1;" clinica_medica

Evidência:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u admin_clinica -pAdm!n_2026#Clin -e "DELETE FROM Prontuario WHERE id_prontuario = 1;" clinica_me
dica
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1142 (42000) at line 1: DELETE command denied to user 'admin_clinica'@'localhost' for table 'prontuario'
```

4. Prova de armazenamento seguro (root)

Teste 4.1

mysql -u root -p -e "SELECT login, senha_hash, salt FROM Usuario_App;" clinica_medica

Evidência:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p -e "SELECT login, senha_hash, salt FROM Usuario_App;" clinica_medica
Enter password: ********
+--------------+------------------------------------------------------------------+----------------------------------+
| login        | senha_hash                                                       | salt                             |
+--------------+------------------------------------------------------------------+----------------------------------+
| ana.recepcao | 8e027d93f719c3353c5d87fa7d4a1f7210ade7de1e1555f68ac6a0c5eec5c684 | 1aee049f5a448784ea663582bdc472c1 |
| dr.carlos    | 42b885060e1ba9eb42cffa99f21f31e67bcdf3d8e54df8217e4933f0bc86d341 | 31d7e3ba507cfd127741ca9ff48a1797 |
| admin.master | 7caebd1a6c70245c4dcf7be27f7ff9a3800856b9fe74ccd926ef65e54721109b | 861e814d8e356a0592c792c7b01cf05b |
+--------------+------------------------------------------------------------------+----------------------------------+
```

Teste 4.2

mysql -u root -p -e "SELECT COUNT(DISTINCT salt) AS salts_unicos, COUNT(*) AS total FROM Usuario_App;" clinica_medica

Evidência:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p -e "SELECT COUNT(DISTINCT salt) AS salts_unicos, COUNT(*) AS total FROM Usuario_App;" c
linica_medica
Enter password: ********
+--------------+-------+
| salts_unicos | total |
+--------------+-------+
|            3 |     3 |
+--------------+-------+
```

Teste 4.3

mysql -u root -p -e "SELECT login, (senha_hash = 'SenhaAna2026!') AS igual_ana, (senha_hash = 'SenhaCarlos2026!') AS igual_carlos, (senha_hash = 'SenhaAdmin2026!') AS igual_admin FROM Usuario_App;" clinica_medica

Evidência:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p -e "SELECT login, (senha_hash = 'SenhaAna2026!') AS igual_ana, (senha_hash = 'SenhaCarl
os2026!') AS igual_carlos, (senha_hash = 'SenhaAdmin2026!') AS igual_admin FROM Usuario_App;" clinica_medica
Enter password: ********
+--------------+-----------+-------------+-------------+
| login        | igual_ana | igual_carlos | igual_admin |
+--------------+-----------+-------------+-------------+
| ana.recepcao |         0 |           0 |           0 |
| dr.carlos    |         0 |           0 |           0 |
| admin.master |         0 |           0 |           0 |
+--------------+-----------+-------------+-------------+
```

Teste 4.4

mysql -u root -p -e "SELECT login, LENGTH(senha_hash) AS tam_hash, LENGTH(salt) AS tam_salt FROM Usuario_App;" clinica_medica

Evidência:

